



## Overestimating vs underestimating

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- 1 \_\_\_\_\_ are the digits in a number that contribute to the accuracy of the number. The first significant figure is the first non-zero digit. Tick 1 correct answer

- ☐ Estimated numbers
- ☐ Important number
- ☐ Real numbers
- ☒ Significant figures

- 2 Which of the following are good, quick estimate calculations for  $577 \div 19.57$ ? Tick 2 correct answers

- ☒  $580 \div 20$
- ☒  $600 \div 20$
- ☐  $580 \div 19.5$
- ☐  $600 \div 19.5$

- 3 Which of the following are a correct estimate for  $\frac{11.239 \times 3.12}{0.488}$ ? Tick 2 correct answers

- ☐  $\frac{11 \times 3}{0.5} = 16.5$
- ☐  $\frac{10 \times 3}{0.5} = 15$
- ☒  $\frac{11 \times 3}{0.5} = 66$
- ☒  $\frac{10 \times 3}{0.5} = 60$



4 Match each calculation to its approximate answer. Write the correct letter in each box

|   |                      |
|---|----------------------|
| a | $\frac{9.8}{0.120}$  |
| b | $\frac{98}{0.111}$   |
| c | $\frac{9.8}{0.2003}$ |
| d | $\frac{980}{0.199}$  |
| e | $\frac{980}{0.498}$  |
| f | $\frac{9}{0.51}$     |

|          |      |
|----------|------|
| <i>b</i> | 1000 |
| <i>c</i> | 50   |
| <i>a</i> | 100  |
| <i>d</i> | 5000 |
| <i>e</i> | 2000 |
| <i>f</i> | 18   |

5 A square plot of grass has a length of 7.88 m (2 d.p). Select the correct estimate for the area of the square grass plot. Tick **1** correct answer

- ☒  $8 \times 8 = 64 \text{ m}^2$
- ☐  $8 \times 8 = 16 \text{ m}^2$
- ☐  $8 + 8 + 8 + 8 = 32 \text{ m}^2$
- ☐  $8 + 8 + 8 + 8 = 32 \text{ m}$

6 Izzy thinks of a number. Truncated to 1 s.f., it is 500. Rounded to the nearest 50, it is 600. It is multiple of 5 and a palindrome . All of its digits are prime numbers. Izzy's number is 575 Fill in the blank

